



GURU TEGH BAHADUR PUBLIC SCHOOL, DURGAPUR

UNIT TEST – II (2026-27) : MARKING SCHEME

CLASS : XI

MAXIMUM MARKS : 50

SUBJECT : COMPUTER SCIENCE (083) TIME ALLOWED : 1 Hr. 45 Min.

General Instructions:

1. Award full marks for any correct alternative logic, code or wording.
2. Deduct $\frac{1}{2}$ mark for minor syntax errors in a program; do not deduct for different variable names.
3. For output questions, the exact output (including brackets, quotes and spaces) is required for full marks.
4. For truth tables, award marks column-wise if a few rows are wrong.

SECTION - A																																	
(a)	XOR (Exclusive OR) gate	1																															
(b)	A + B	1																															
(c)	all its inputs are 1	1																															
(d)	lcomme T	1																															
(e)	ce lo	1																															
(f)	#THONPROGRAM	1																															
(g)	['I', 'ter', 'atio', 'al']	1																															
(h)	['TAC', 'TIC']	1																															
(i)	remove()	1																															
(j)	Error – ValueError: substring not found (the string has no uppercase "C")	1																															
(k)	pass	1																															
(l)	5 times (i = 2, 6, 10, 14, 18)	1																															
(m)	False	1																															
SECTION - B																																	
Q.2	<table><tr><td>A</td><td>B</td><td>A + B</td><td>(A + B)'</td><td>A'</td><td>B'</td><td>A'·B'</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr></table>	A	B	A + B	(A + B)'	A'	B'	A'·B'	0	0	0	1	1	1	0	1	1	0	1	0	1	0	1	0	0	0	1	1	1	0	0	0	(2)
	A	B	A + B	(A + B)'	A'	B'	A'·B'																										
	0	0	0	1	1	1																											
	0	1	1	0	1	0																											
	1	0	1	0	0	0																											
	1	1	1	0	0	0																											
Columns (A + B)' and A'·B' are identical, hence proved. (Correct table 1½, conclusion ½)																																	

Q.3	(a) <code>review.find("good")</code> (or <code>review.index("good")</code>) (b) <code>L1.sort(reverse=True)</code>	(1+1)															
Q.4	<code>[10, 20, 25, 30, 40, 5, 15]</code> 3 1 (1 mark for the list, ½ each for 3 and 1)	(2)															
SECTION - C																	
Q.5	(a) $A \oplus B = A'B + AB'$ (b) Truth table: <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>$A \oplus B$</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table> (c) Circuit: $A \rightarrow \text{NOT gate} \rightarrow A'$; $B \rightarrow \text{NOT gate} \rightarrow B'$. AND gate 1 takes A' and B ; AND gate 2 takes A and B' . The outputs of the two AND gates go into an OR gate, whose output is $A'B + AB'$.	A	B	$A \oplus B$	0	0	0	0	1	1	1	0	1	1	1	0	(1+1+1)
A	B	$A \oplus B$															
0	0	0															
0	1	1															
1	0	1															
1	1	0															
Q.6	QP^{-14} $S \rightarrow \text{next char 'Q'}; Q \rightarrow \text{next char 'P'}; P \rightarrow \text{next char '-'}; '-' \rightarrow '^'; '2' \rightarrow 1; '5' \rightarrow 4.$ (½ mark per character)	(3)															
Q.7	<pre> n = int(input("How many numbers? ")) L = [] for i in range(n): L.append(float(input("Enter number: "))) large = small = L[0] total = 0 for x in L: if x > large: large = x if x < small: small = x total += x print("Largest:", large) print("Smallest:", small) print("Mean:", total / n) </pre> (Input into list 1, largest/smallest logic 1, mean ½, display ½)	(3)															

SECTION - D																																				
Q.8	(a) $(A + B)(A + C)$ $= AA + AC + AB + BC$ (distributive law) $= A + AC + AB + BC$ ($AA = A$) $= A(1 + C + B) + BC$ $= A + BC$ ($1 + X = 1$) (b) <table><tr><td>A</td><td>B</td><td>A'</td><td>A'B</td><td>A + A'B</td><td>A + B</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table> The last two columns are identical, hence verified.					A	B	A'	A'B	A + A'B	A + B	0	0	1	0	0	0	0	1	1	1	1	1	1	0	0	0	1	1	1	1	0	0	1	1	(2+2)
A	B	A'	A'B	A + A'B	A + B																															
0	0	1	0	0	0																															
0	1	1	1	1	1																															
1	0	0	0	1	1																															
1	1	0	0	1	1																															
Q.9	<pre>s = input("Enter a string: ") if s.lower() == s.lower()[::-1]: print("Palindrome") else: print("Not a palindrome") count = 0 for w in s.split(): if w[0] in "aeiouAEIOU": count += 1 print("Words beginning with a vowel:", count)</pre> (Accept a loop-based palindrome check.)					(2+2)																														
Q.10	(a) Year . 0. at All the best (b) ceieP0 (c) ['Watersk', '', 'ng ', 's thr', 'll', 'ng!'] (d) It					(1×4)																														
SECTION - E																																				
Q.11	<pre>n = int(input("How many elements? ")) L = [] for i in range(n): L.append(int(input("Enter element: "))) U = [] for x in L: if x not in U: U.append(x) print("List without duplicates:", U) key = int(input("Enter number to search: ")) for i in range(len(L)): if L[i] == key: print("Found at position", i + 1) break else: print("Not found")</pre> (a) 2 marks (b) 3 marks. Accept index-based or 0-based positions if stated.					(2+3)																														

Q.12	(a) ['K', 'R'] (Kaziranga ends in 'a', Ranthambhore ends in 'e') (b) <pre>n = int(input("Enter a positive integer: ")) s = 0 rev = 0 while n > 0: d = n % 10 s += d rev = rev * 10 + d n //= 10 print("Sum =", s, ", Reverse =", rev)</pre>	(2+3)
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SOURCES OF QUESTIONS TAKEN FROM PREVIOUS PAPERS

Q1 (d)	CBSE CS Additional Practice Paper 2023-24, Q3	
Q1 (e), (f), (g)	CBSE CS Sample Paper 2024-25, Q5, Q2, Q4	
Q1 (h), (j)	CBSE Board Exam 2025 (CS, Q.P. Code 91), Q6 and Q3 – options removed	
Q1 (i)	Adapted from CBSE CS Sample Paper 2024-25, Q8	
Q1 (m)	CBSE CS Sample Paper 2025-26, Q3	
Q3	CBSE CS Sample Paper 2025-26, Q24 (A)	
Q6	CBSE CS Sample Paper 2025-26, Q31 (A)	
Q10 (a)	CBSE CS Sample Paper 2022-23, Q5	
Q10 (b)	CBSE CS Sample Paper 2023-24, Q8	
Q10 (c), (d)	CBSE CS Sample Paper 2025-26, Q14 and Q2	
Q12 (a)	CBSE CS Sample Paper 2025-26, Q31 (B)	

***** End of Paper *****